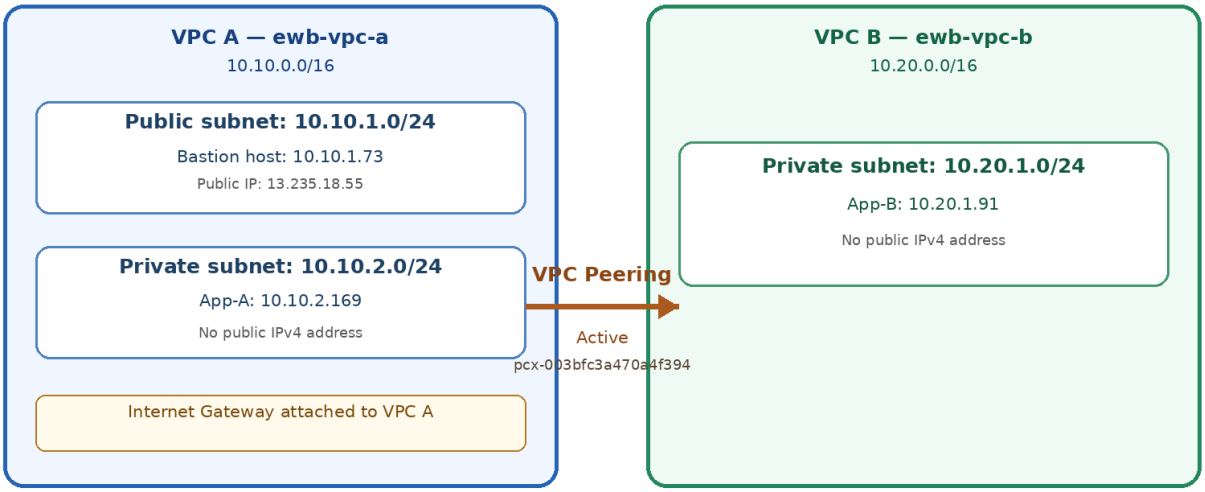
AWS VPC PEERING

Student Project Report

AWS VPC Peering — Project Architecture



Route tables send 10.20.0.0/16 and 10.10.0.0/16 through the peering connection.

SSH administration uses the bastion host; App-A and App-B communicate over private IPs.

Student Name	Sudarshan Birajdar
Project	AWS VPC Peering — EWB Cloud Labs
AWS Region	Asia Pacific (Mumbai) — ap-south-1
Completion Date	28 October 2026

Prepared from the AWS console and terminal screenshots supplied for this project. Account-identifying information has been obscured where visible.

1. Project Overview

The objective of this project was to configure two separate Amazon Virtual Private Clouds (VPCs) and enable private communication between workloads in them using VPC Peering. A bastion host in VPC A provides an administrative entry point, while the two application instances use private IPv4 addresses and do not have public IPv4 addresses.

Project goals

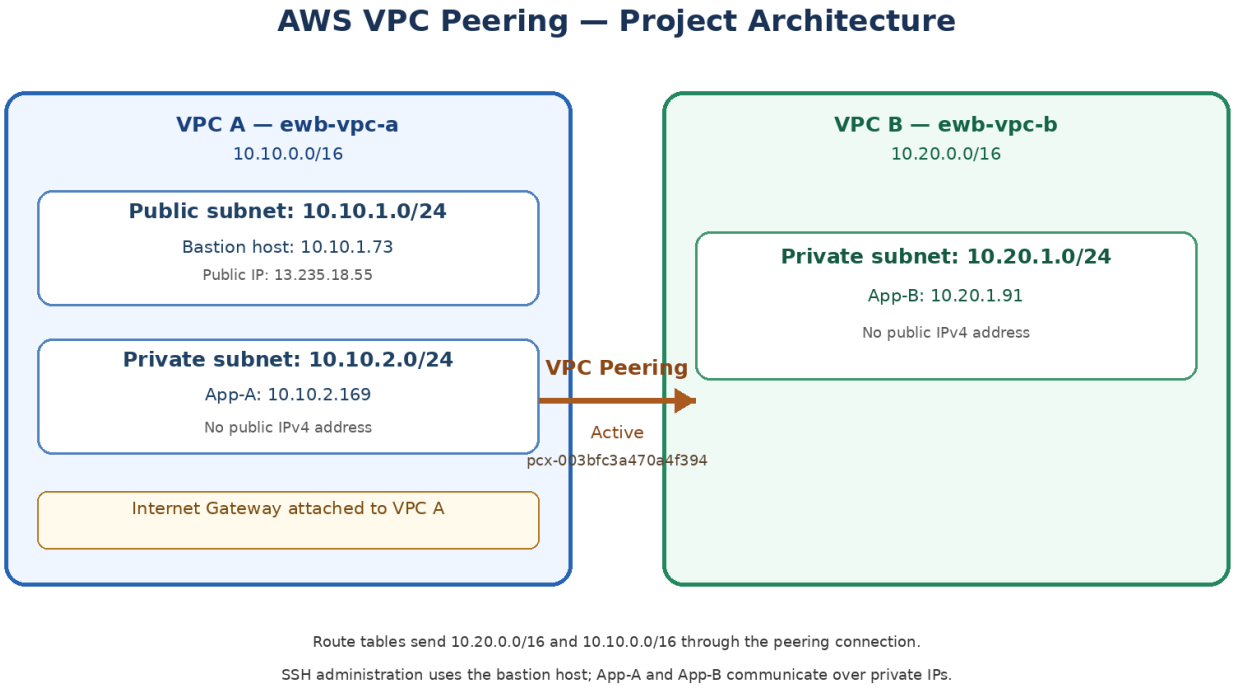
• Create two custom VPCs with non-overlapping CIDR ranges.
• Configure public and private subnets and suitable route tables.
• Attach an Internet Gateway to VPC A for the bastion subnet.
• Establish and verify an active VPC peering connection.
• Apply security-group rules for SSH administration and ICMP connectivity tests.
• Validate private connectivity between App-A and App-B.

2. Network Inventory

Resource	Name / ID	Addressing / Status
VPC A	ewb-vpc-a — vpc-0981b3ce535a4c65c	10.10.0.0/16
VPC B	ewb-vpc-b — vpc-02afc611d2ee005b6	10.20.0.0/16
Public subnet A	ewb-a-public	10.10.1.0/24
Private subnet A	ewb-a-private	10.10.2.0/24
Private subnet B	ewb-b-private	10.20.1.0/24
Bastion EC2	ewb-bastion	10.10.1.73; public IPv4 13.235.18.55
App-A EC2	ewb-app-a	10.10.2.169; no public IPv4
App-B EC2	ewb-app-b	10.20.1.91; no public IPv4
Peering	ewb-a-b-peer — pcx-003bfc3a470a4f394	Active
Internet Gateway	ewb-igw-a — igw-0e76575671134477e	Attached to VPC A

Note: The supplied screenshots show the three project subnets and the VPCs in the Mumbai region. The VPC peering details confirm non-overlapping CIDRs and Active status.

3. Architecture and Routing Design



VPC peering creates private network connectivity between the two VPCs without requiring traffic to traverse the public internet. Peering does not automatically configure routes, so each relevant route table must contain a route to the other VPC CIDR with the peering connection as the target.

Routing configuration evidenced in the screenshots

• VPC A public route table: 0.0.0.0/0 points to the Internet Gateway; 10.10.0.0/16 is the local route; 10.20.0.0/16 points to pcx-003bfc3a470a4f394.
• VPC A private route table: the 10.20.0.0/16 destination is routed through the peering connection; the local route remains for VPC A.
• VPC B private route table: 10.10.0.0/16 points to the peering connection; 10.20.0.0/16 is the local route.

Security design

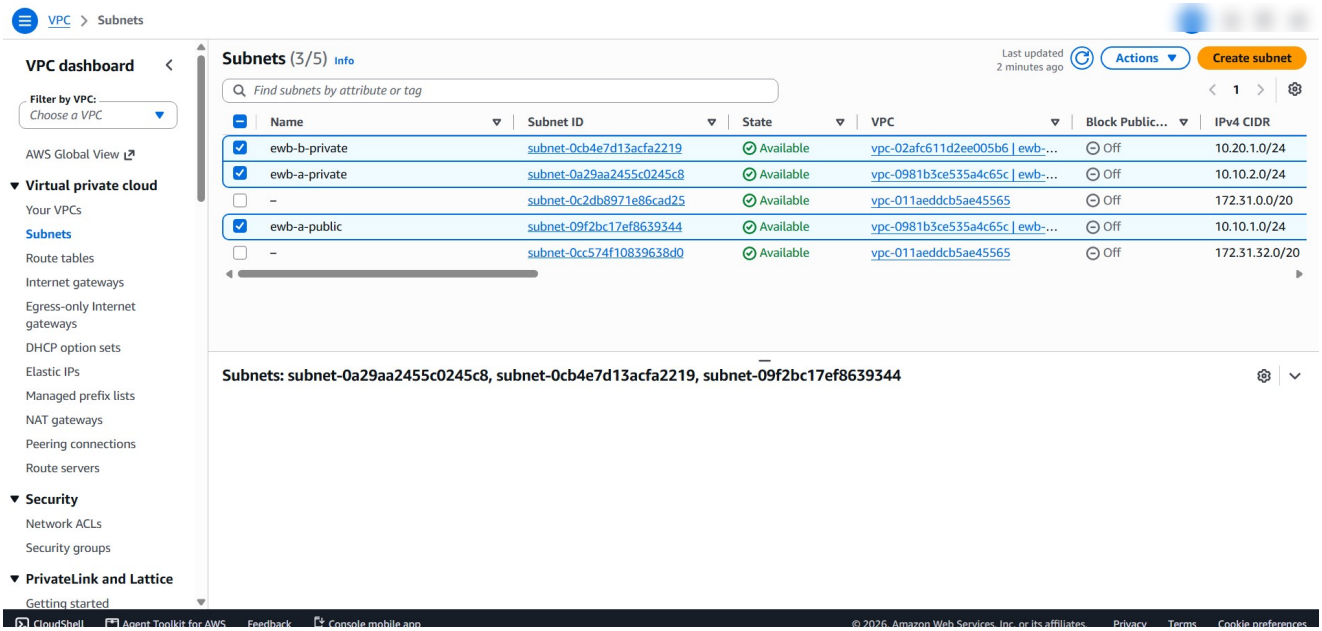
The bastion security group allows SSH (TCP 22) from the administrator's permitted source. The application security groups are intended to allow SSH from the bastion private address and ICMP traffic from the opposite application instance for testing. Security groups are stateful; route-table paths and security-group rules must both allow the intended communication.

4. Implementation Evidence

The following screenshots are arranged by configuration area. Personal/account-identifying details in the AWS console header and visible account-owner fields have been obscured in the report copies. Network IDs, private IPs, CIDR ranges, route targets, and test output remain readable.

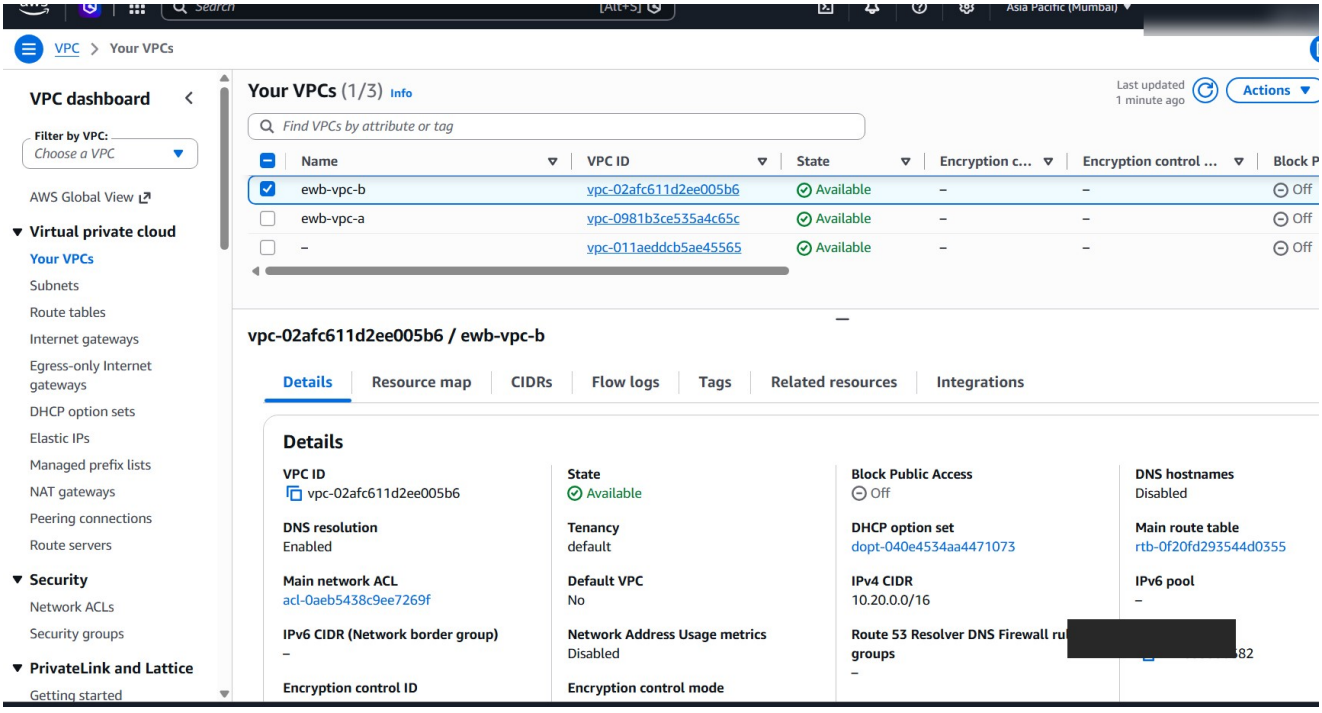
4.1 Subnet configuration

The screenshot lists the three project subnets: ewb-a-public (10.10.1.0/24), ewb-a-private (10.10.2.0/24), and ewb-b-private (10.20.1.0/24).



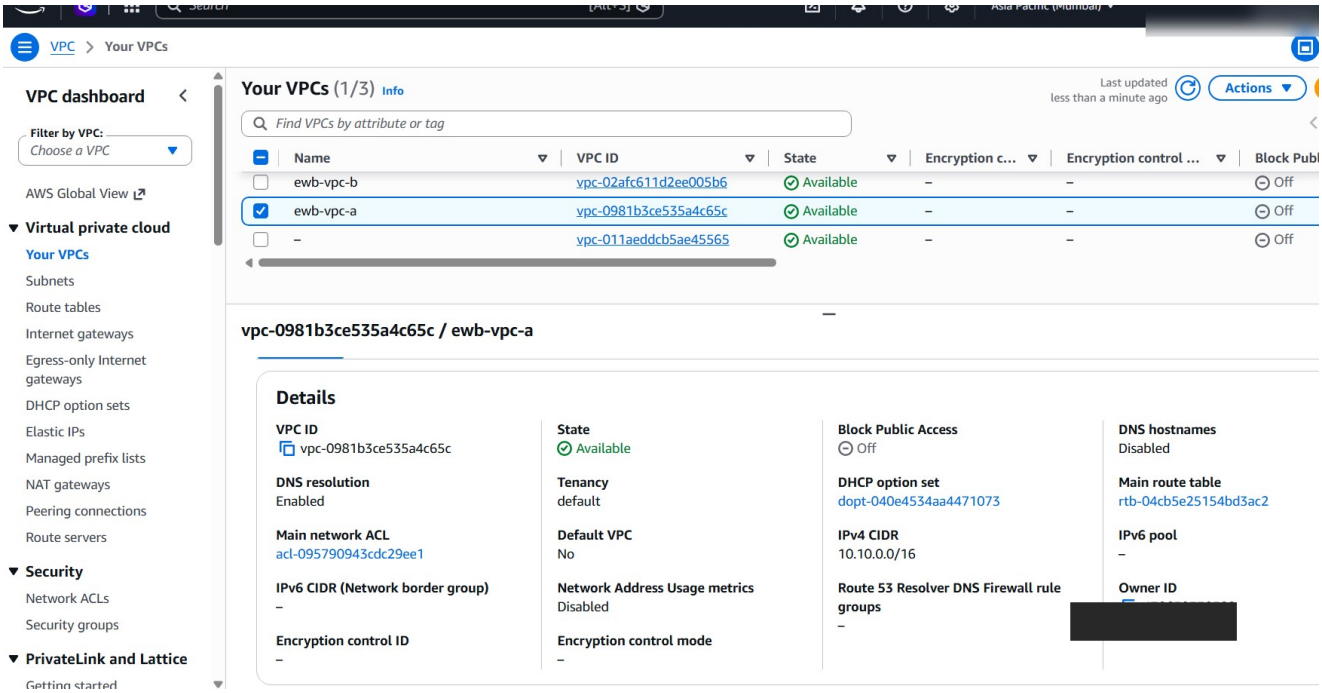
4.2 VPC B details

Confirms ewb-vpc-b is available and uses 10.20.0.0/16.



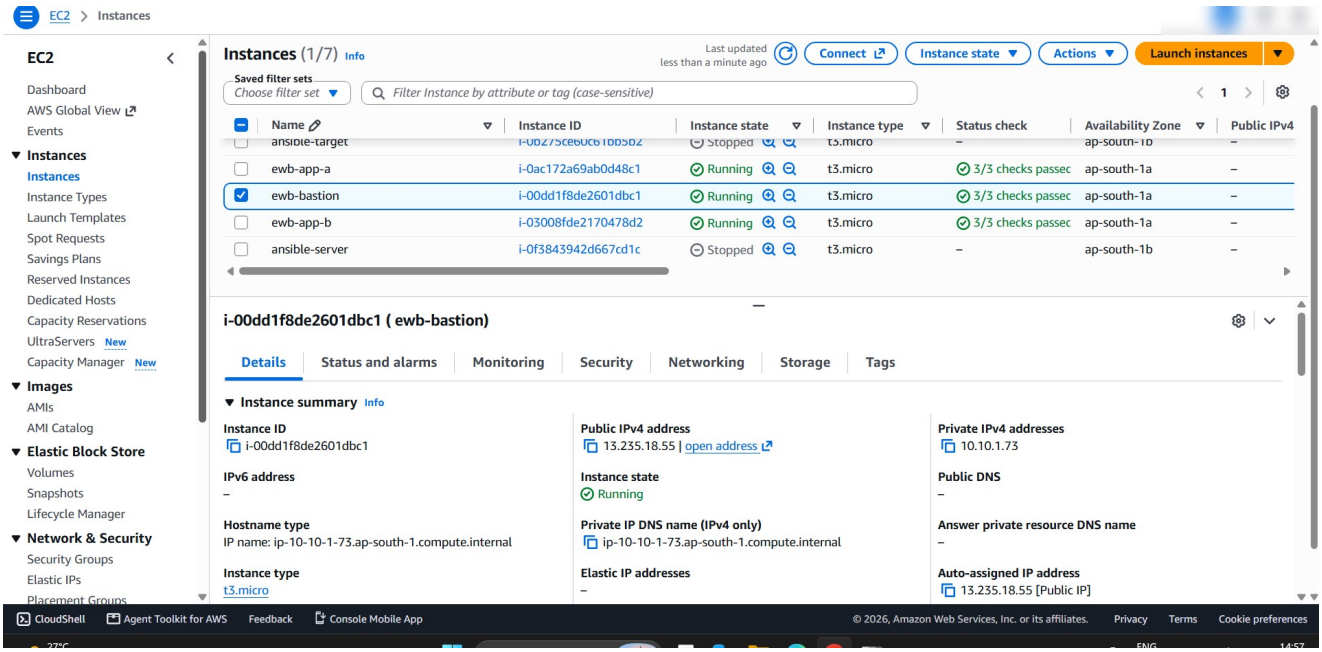
4.3 VPC A details

Confirms ewb-vpc-a is available and uses 10.10.0.0/16.


VPC > Your VPCs
VPC dashboard
Filter by VPC:
Choose a VPC
AWS Global View
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only Internet gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers
Security
Network ACLs
Security groups
PrivateLink and Lattice
Getting started
Your VPCs (1/3) Info
Last updated less than a minute ago
Actions
Find VPCs by attribute or tag
Name
VPC ID
State
Encryption c...
Encryption control ...
Block Publ
ewb-vpc-b
vpc-02afc611d2ee005b6
Available
Off
ewb-vpc-a
vpc-0981b3ce535a4c65c
Available
Off
vpc-011aeddcb5ae45565
Available
Off
vpc-0981b3ce535a4c65c / ewb-vpc-a
Details
VPC ID
vpc-0981b3ce535a4c65c
State
Available
Block Public Access
Off
DNS hostnames
Disabled
DNS resolution
Enabled
Tenancy
default
DHCP option set
dopt-040e4534aa4471073
Main route table
rtb-04cb5e25154bd3ac2
Main network ACL
acl-095790943cdc29ee1
Default VPC
No
IPv4 CIDR
10.10.0.0/16
IPv6 pool
IPv6 CIDR (Network border group)
Network Address Usage metrics
Disabled
Route 53 Resolver DNS Firewall rule groups
Owner ID
Encryption control ID
Encryption control mode

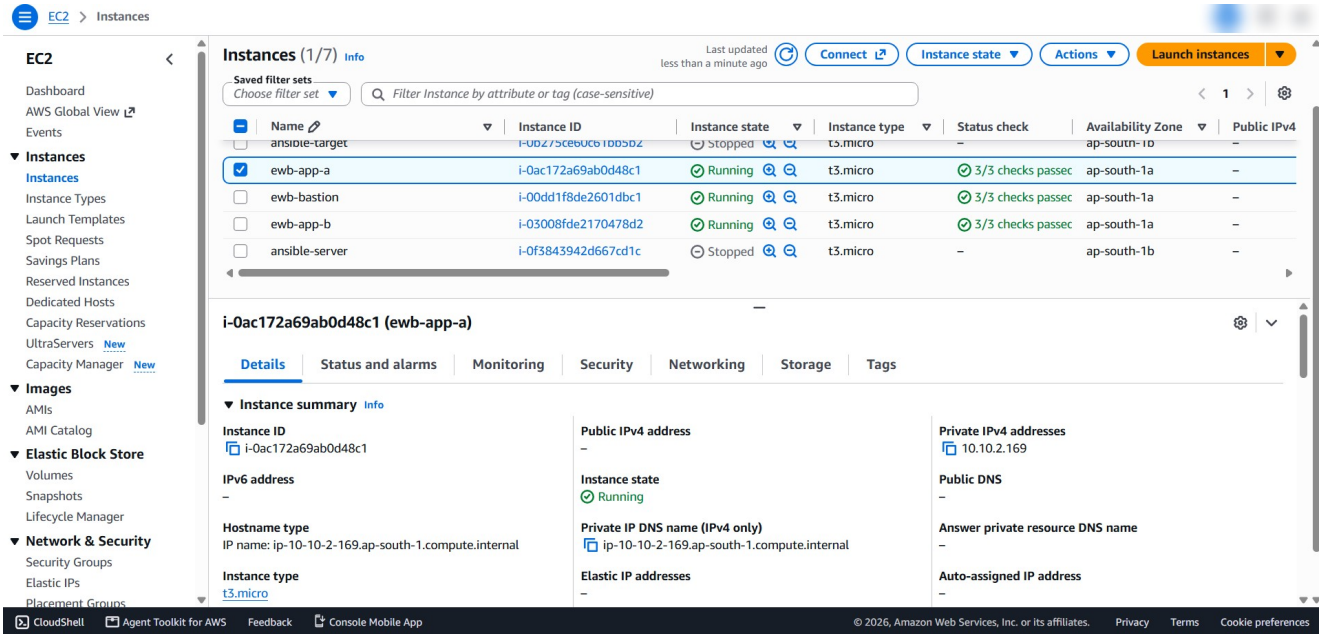
4.4 Bastion instance details

Shows ewb-bastion running with private IP 10.10.1.73 and public IPv4 address. The public IP is retained as implementation evidence.



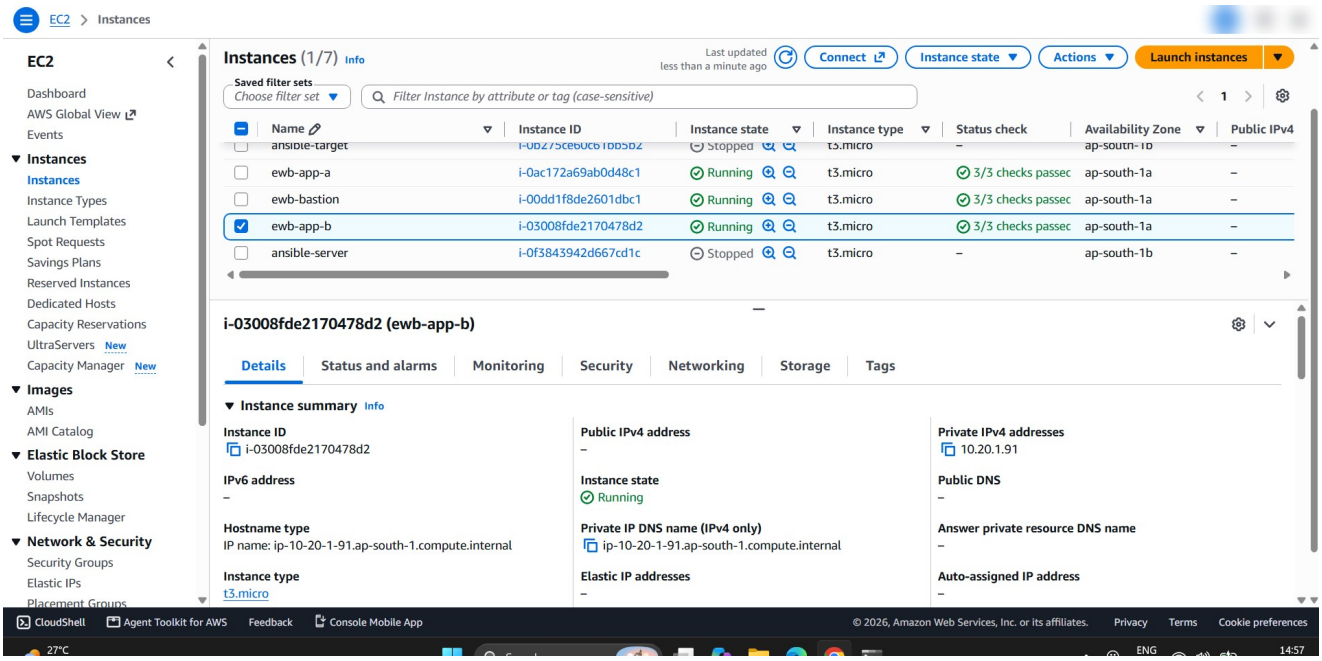
4.5 App-A instance details

Shows ewb-app-a running with private IP 10.10.2.169 and no public IPv4 address.



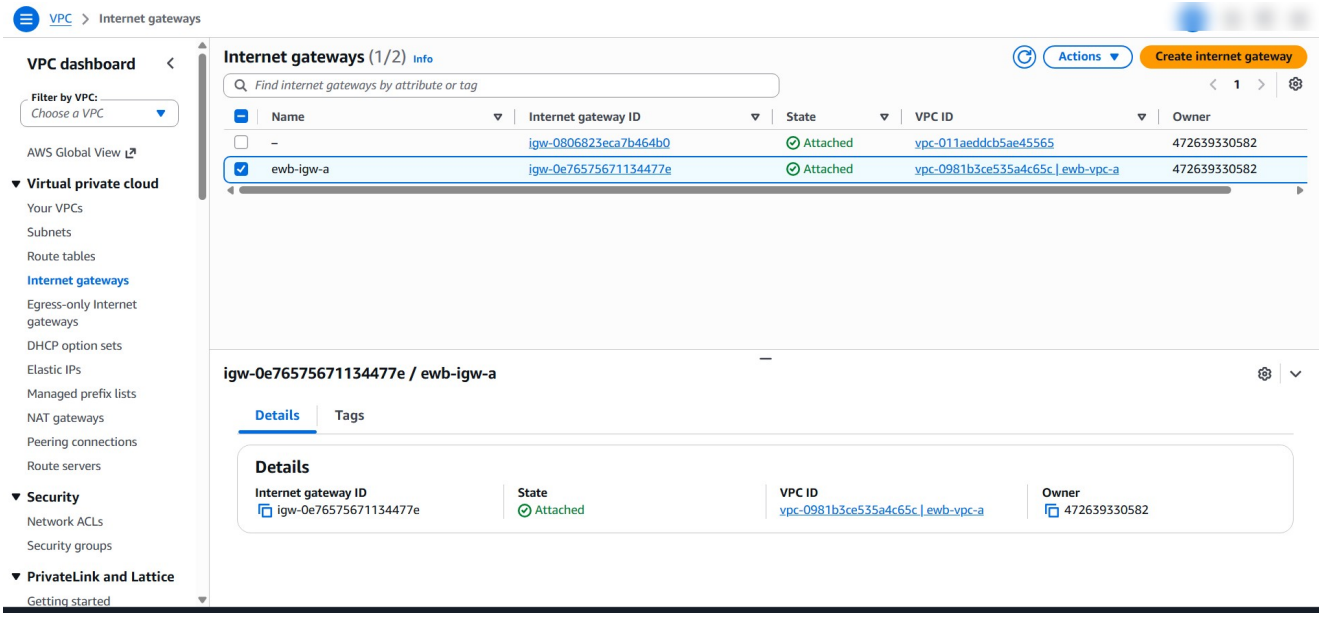
4.6 App-B instance details

Shows ewb-app-b running with private IP 10.20.1.91 and no public IPv4 address.


EC2 > Instances
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Network & Security
Security Groups
Elastic IPs
Placement Groups
Instances (1/7) Info
Last updated less than a minute ago
Connect
Instance state
Actions
Launch instances
Saved filter sets
Choose filter set
Filter Instance by attribute or tag (case-sensitive)
Name
Instance ID
Instance state
Instance type
Status check
Availability Zone
Public IPv4
ansible-target
i-0b275ce6dc61bb5b2
Stopped
t3.micro
ap-south-1b
ewb-app-a
i-0ac172a69ab0d48c1
Running
t3.micro
3/3 checks passed
ap-south-1a
ewb-bastion
i-00dd1f8de2601dbc1
Running
t3.micro
3/3 checks passed
ap-south-1a
ewb-app-b
i-03008fde2170478d2
Running
t3.micro
3/3 checks passed
ap-south-1a
ansible-server
i-0f3843942d667cd1c
Stopped
t3.micro
ap-south-1b
i-03008fde2170478d2 (ewb-app-b)
Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags
Instance summary Info
Instance ID
i-03008fde2170478d2
Public IPv4 address
Private IPv4 addresses
10.20.1.91
IPv6 address
Instance state
Running
Public DNS
Hostname type
IP name: ip-10-20-1-91.ap-south-1.compute.internal
Private IP DNS name (IPv4 only)
ip-10-20-1-91.ap-south-1.compute.internal
Answer private resource DNS name
Instance type
t3.micro
Elastic IP addresses
Auto-assigned IP address
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences

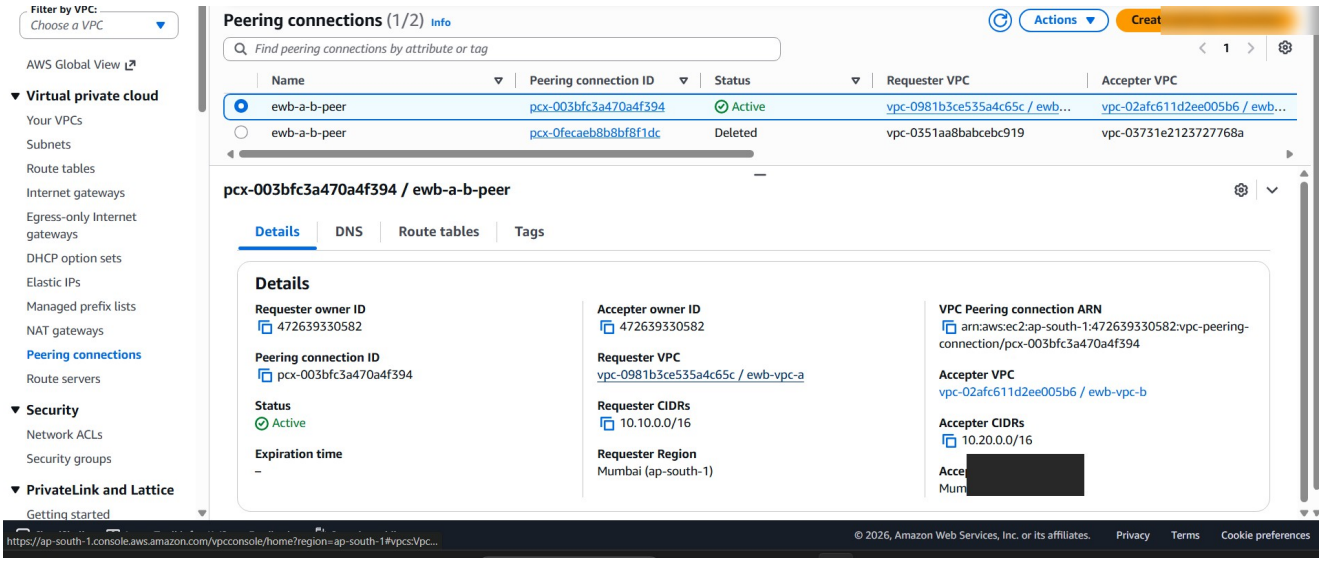
4.7 Internet Gateway

Shows ewb-igw-a attached to VPC A.



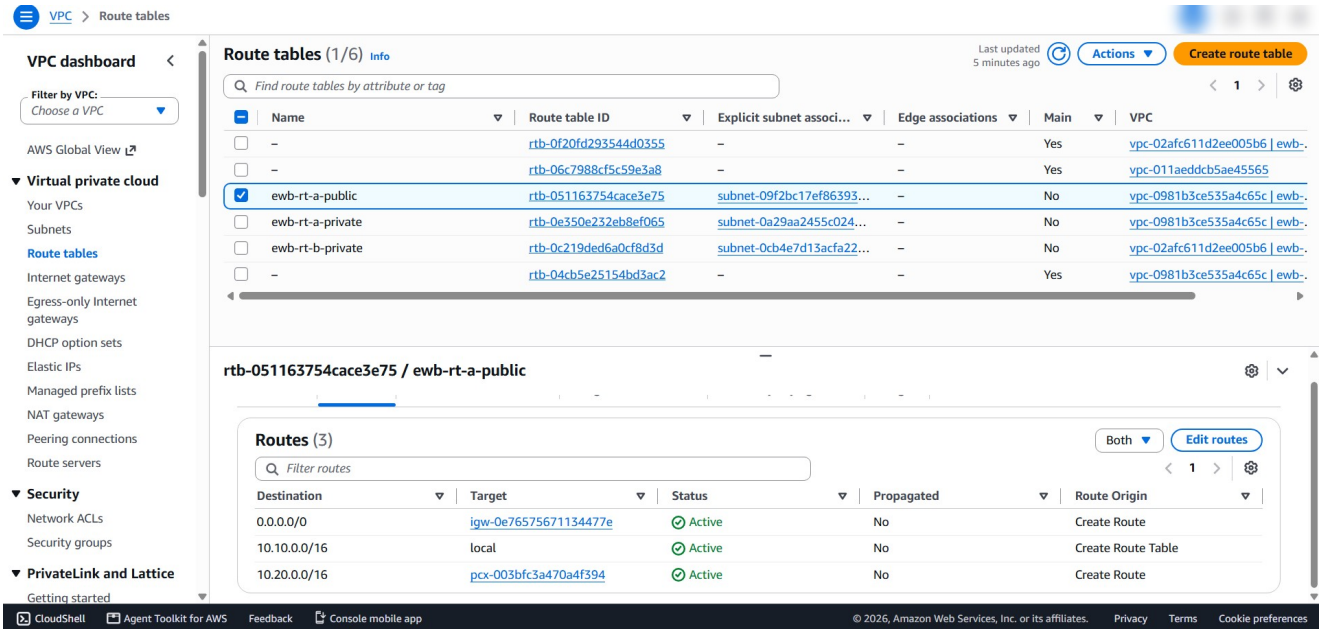
4.8 VPC peering connection

Shows ewb-a-b-peer (pcx-003bfc3a470a4f394) in Active status, with VPC A and VPC B as requester and accepter.



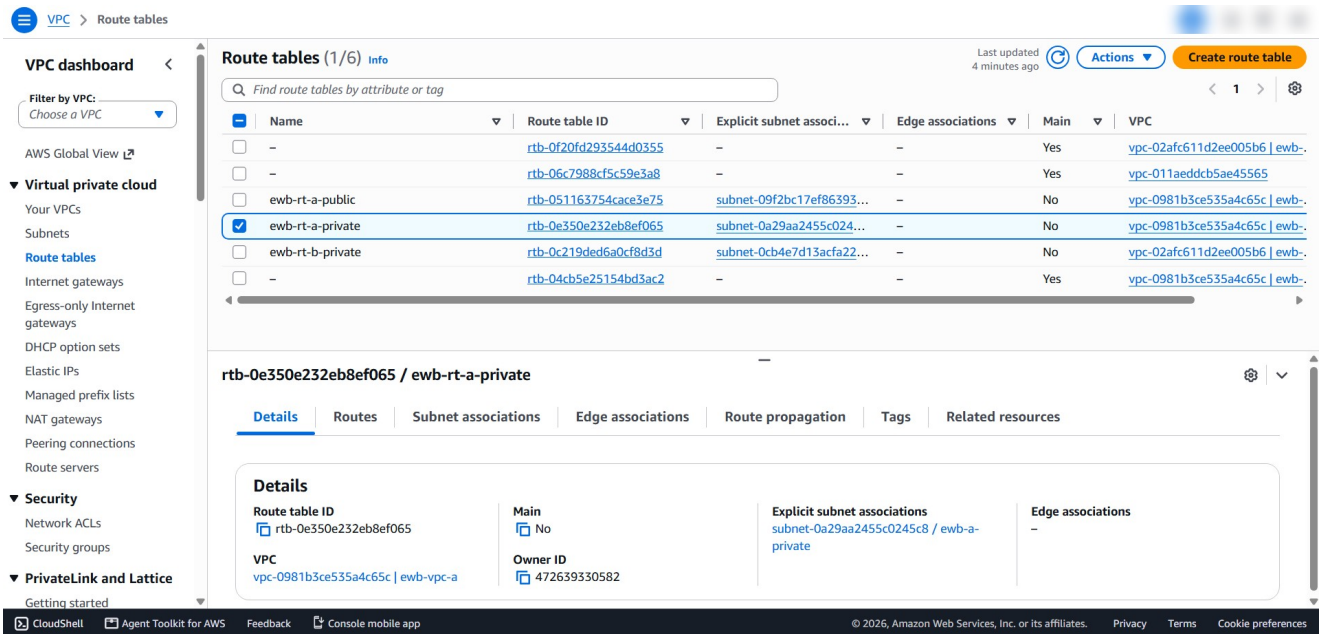
4.9 VPC A public route table

Shows the default Internet Gateway route, VPC A local route, and route to VPC B through the peering connection.


VPC > Route tables
VPC dashboard
Filter by VPC:
Choose a VPC
AWS Global View
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only Internet gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers
Security
Network ACLs
Security groups
PrivateLink and Lattice
Getting started
Route tables (1/6) Info
Last updated 5 minutes ago
Actions
Create route table
Find route tables by attribute or tag
Name
Route table ID
Explicit subnet associ...
Edge associations
Main
VPC
– rtb-0f20fd293544d0355 – – Yes vpc-02afc611d2ee005b6 | ewb-.
– rtb-06c7988cf5c59e3a8 – – Yes vpc-011aeddcb5ae45565
ewb-rt-a-public rtb-051163754cace3e75 subnet-09f2bc17ef86393... – No vpc-0981b3ce535a4c65c | ewb-.
ewb-rt-a-private rtb-0e350e232eb8ef065 subnet-0a29aa2455c024... – No vpc-0981b3ce535a4c65c | ewb-.
ewb-rt-b-private rtb-0c219ded6a0cf8d3d subnet-0cb4e7d13acfa22... – No vpc-02afc611d2ee005b6 | ewb-.
– rtb-04cb5e25154bd3ac2 – – Yes vpc-0981b3ce535a4c65c | ewb-.
rtb-051163754cace3e75 / ewb-rt-a-public
Routes (3)
Both
Edit routes
Filter routes
Destination
Target
Status
Propagated
Route Origin
0.0.0.0/0 igw-0e76575671134477e Active No Create Route
10.10.0.0/16 local Active No Create Route Table
10.20.0.0/16 pcx-003bfc3a470a4f394 Active No Create Route
CloudShell
Agent Toolkit for AWS
Feedback
Console mobile app
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences

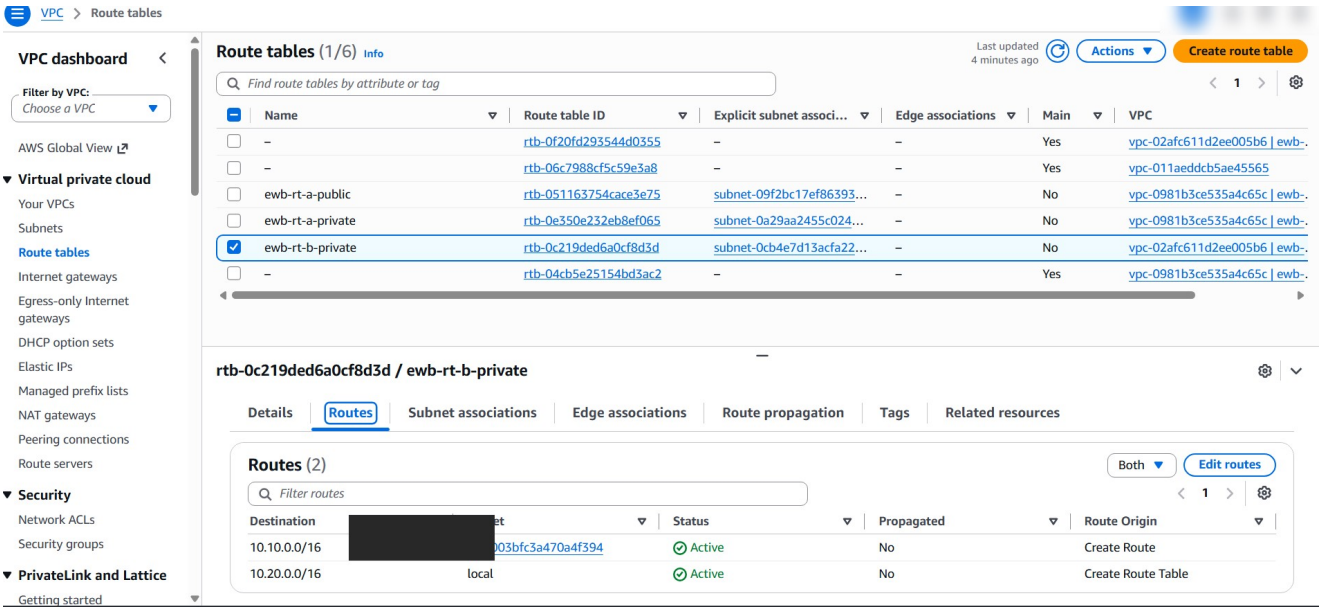
4.10 VPC A private route table association

Shows ewb-rt-a-private associated with the ewb-a-private subnet.



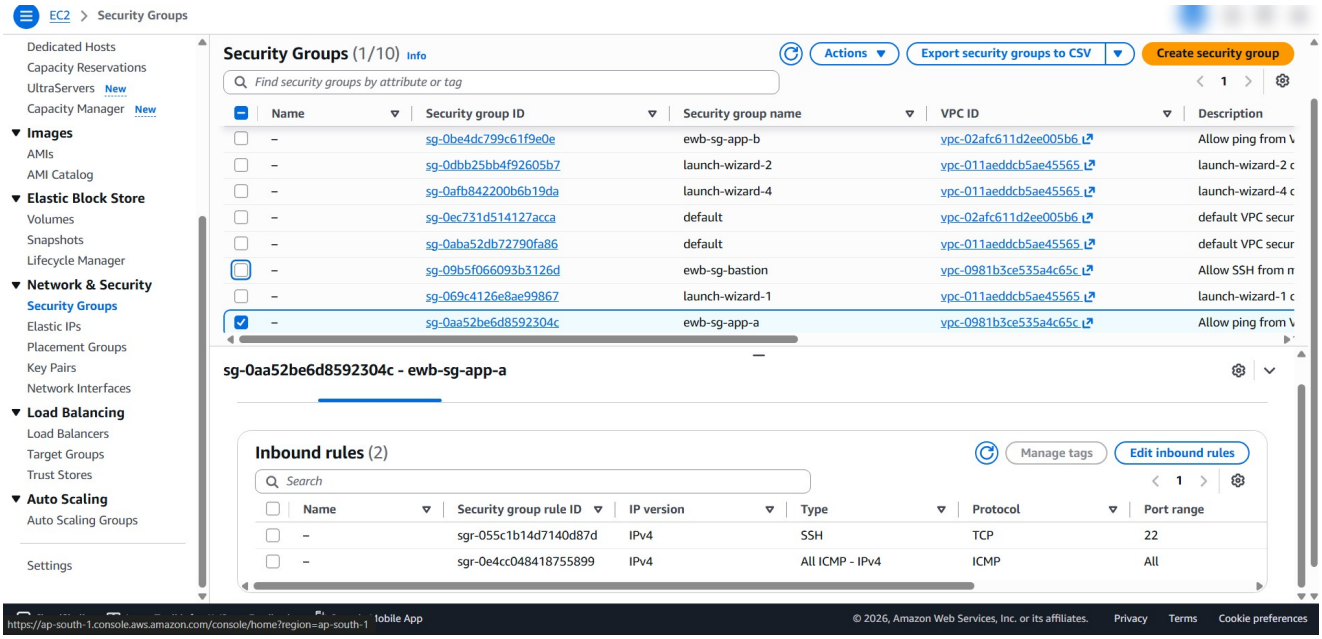
4.11 VPC B private route table

Shows the route from VPC B to 10.10.0.0/16 through the peering connection.



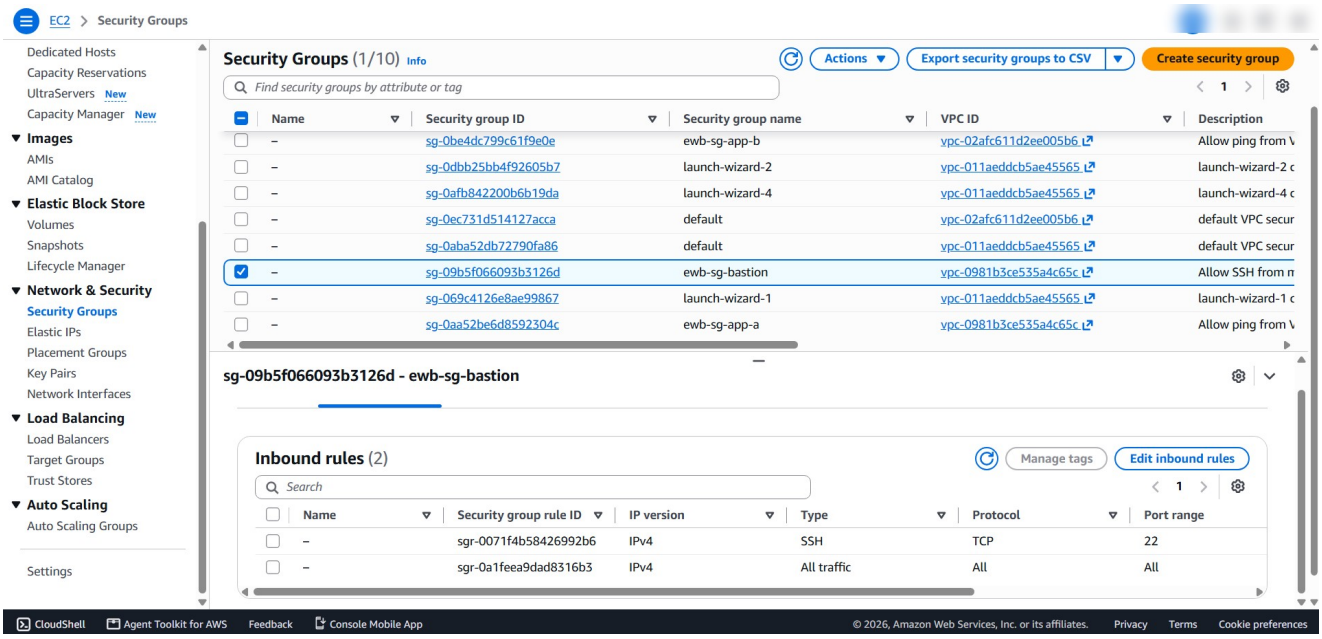
4.12 App-A security group

Shows inbound SSH and ICMP rule types. The source CIDRs are not fully visible in this screenshot.


EC2 > Security Groups
Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Network & Security
Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces
Load Balancing
Load Balancers
Target Groups
Trust Stores
Auto Scaling
Auto Scaling Groups
Settings
Security Groups (1/10) Info
Actions
Export security groups to CSV
Create security group
Find security groups by attribute or tag
Name
Security group ID
Security group name
VPC ID
Description
sg-0be4dc799c61f9e0e ewb-sg-app-b
sg-0dbb25bb4f92605b7 launch-wizard-2
sg-0afb842200b6b19da launch-wizard-4
sg-0ec731d514127acca default
sg-0aba52db72790fa86 default
sg-09b5f066093b3126d ewb-sg-bastion
sg-069c4126e8ae99867 launch-wizard-1
sg-0aa52be6d8592304c ewb-sg-app-a
sg-0aa52be6d8592304c - ewb-sg-app-a
Inbound rules (2)
Manage tags
Edit inbound rules
Search
Name
Security group rule ID
IP version
Type
Protocol
Port range
sgr-055c1b14d7140d87d IPv4 SSH TCP 22
sgr-0e4cc048418755899 IPv4 All ICMP - IPv4 ICMP All
https://ap-south-1.console.aws.amazon.com/console/home?region=ap-south-1
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences

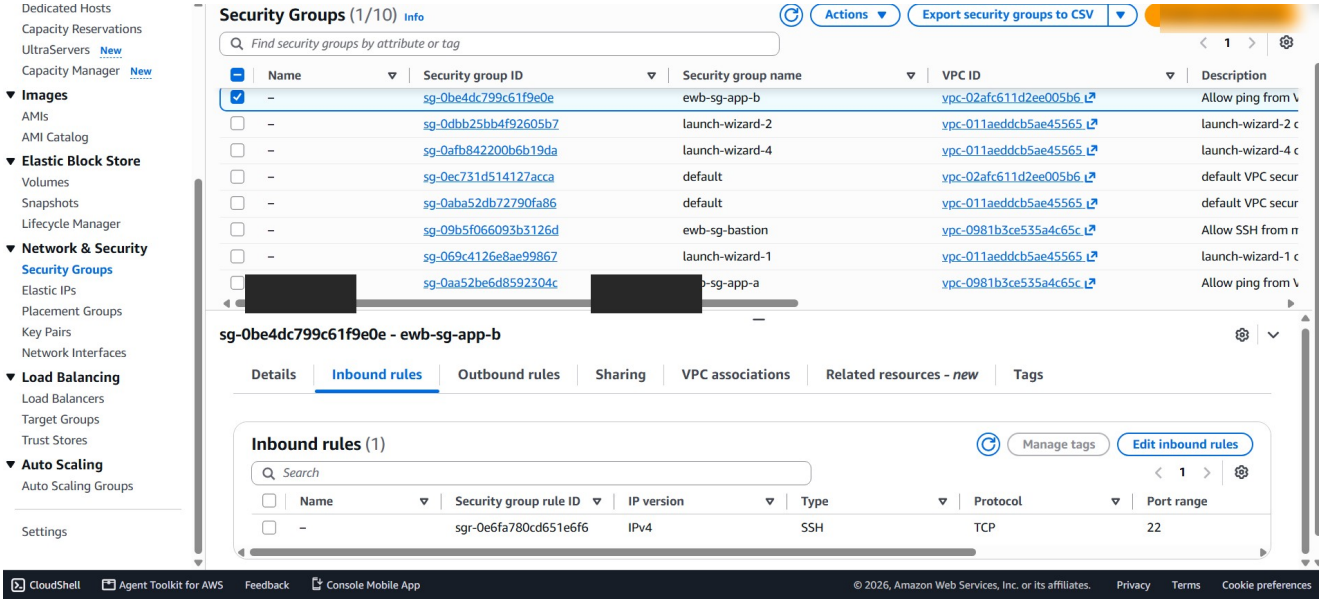
4.13 Bastion security group

Shows SSH and an all-traffic outbound/inbound rule entry in the displayed panel. Review the actual inbound source before final submission; avoid exposing SSH to all IPv4 addresses.



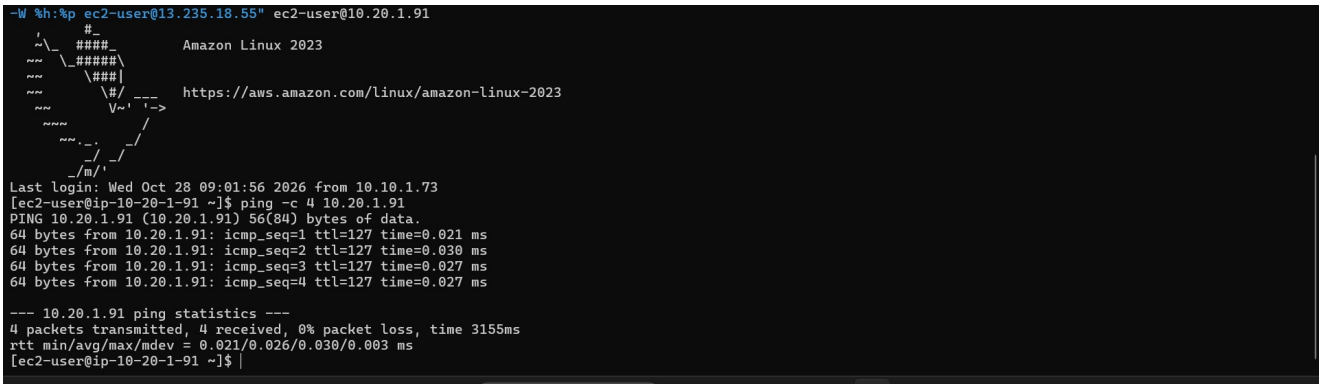
4.14 App-B security group

Shows SSH inbound rule. The screenshot does not show the ICMP rule, so use a separate screenshot of the final restored rules if required.



4.15 App-B terminal ping test

From App-B, ping to 10.20.1.91 returned 4/4 packets with 0% packet loss. This proves App-B can reach its own address, not the cross-VPC path; include a cross-VPC ping screenshot from App-B to App-A for stronger evidence.


-W %h:%p ec2-user@13.235.18.55" ec2-user@10.20.1.91
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
Last login: Wed Oct 28 09:01:56 2026 from 10.10.1.73
[ec2-user@ip-10-20-1-91 ~]$ ping -c 4 10.20.1.91
PING 10.20.1.91 (10.20.1.91) 56(84) bytes of data.
64 bytes from 10.20.1.91: icmp_seq=1 ttl=127 time=0.021 ms
64 bytes from 10.20.1.91: icmp_seq=2 ttl=127 time=0.030 ms
64 bytes from 10.20.1.91: icmp_seq=3 ttl=127 time=0.027 ms
64 bytes from 10.20.1.91: icmp_seq=4 ttl=127 time=0.027 ms
--- 10.20.1.91 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3155ms
rtt min/avg/max/mdev = 0.021/0.026/0.030/0.003 ms
[ec2-user@ip-10-20-1-91 ~]$

5. Validation Results and Limitations

The supplied evidence supports successful creation of the two VPCs, subnet layout, Internet Gateway attachment, active peering connection, and relevant route-table entries. The EC2 console screenshots show the three project instances running and confirm that App-A and App-B have no public IPv4 addresses.

Test evidence available

• The terminal screenshot shows 4 packets transmitted and 4 received with 0% packet loss when pinging 10.20.1.91 from App-B. Because the destination matches App-B's own private IP, this is a local-interface test and does not independently prove cross-VPC connectivity.
• The conversation context records successful cross-VPC ping tests in both directions earlier, but those exact cross-VPC terminal screenshots were not included in this upload batch. They should be added to the final evidence set if the evaluator requires screenshot proof.

Evidence still needed for a fully complete report

1. Cross-VPC ping from App-A (10.10.2.169) to App-B (10.20.1.91), showing 4/4 replies.
2. Cross-VPC ping from App-B (10.20.1.91) to App-A (10.10.2.169), showing 4/4 replies.
3. Controlled failure screenshot after temporarily removing the ICMP inbound rule on App-B's security group, followed by a recovery screenshot after restoring the rule.
4. Cleanup evidence showing project resources were terminated/deleted, or a brief statement that cleanup is pending.

No missing test or cleanup result is presented as completed. Add the above evidence before submitting if the workbook requires all items.

6. Key Learning Outcomes

• VPCs provide isolated virtual networks with configurable IPv4 CIDR ranges.
• Subnets divide a VPC network into smaller ranges and can be designated for public or private workloads based on routing and address configuration.
• Route tables determine where packets are sent. Peering routes must be configured on both sides for two-way connectivity.
• A VPC Peering connection enables private routing between VPCs but does not automatically change route tables or security groups.
• Security groups control permitted inbound and outbound traffic; both the route path and the security rules must allow the traffic.
• A bastion host can provide controlled SSH access to private instances without assigning public IP addresses to those instances.
• Ping and SSH tests help validate network reachability and administration paths, while controlled failure/recovery demonstrates the effect of security-group rules.

7. Cleanup and Cost Control

Cleanup evidence was not present in the screenshots supplied. Before final submission, verify whether the lab requires resource deletion and capture evidence after cleanup. If cleanup is required, terminate the three project EC2 instances, confirm no billable EBS volumes or snapshots remain, delete the peering connection and project routes, remove security groups after dependent interfaces are gone, delete custom route tables and subnets, detach/delete the Internet Gateway, and delete the two project VPCs. Remove the key pair only if it is no longer needed. Do not delete unrelated resources in the account.

8. Conclusion

This project demonstrates the core configuration of VPC peering between two custom AWS VPCs in the Mumbai region. The supplied screenshots confirm the network inventory, attached Internet Gateway, Active peering status, route-table design, EC2 instance private addressing, and security-group configuration. The final submission should include the missing cross-VPC test, controlled failure/recovery, and cleanup evidence to make the validation section complete.

Prepared for EWB Cloud Labs — AWS VPC Peering Student Project